

questions 1,2 consider the current through a 1 mH inductor is $i(t)=60 \cos 100t$ mA.
terminal voltage across the inductor is:

2) The Energy stored in the inductor is:

Prob 2: For questions 3,4,5,6 consider the circuit shown in figure 1. The switch has been closed for a long time. At $t=0$, the switch is opened. Answer:

3) The initial current $i(t=0+)$ is:

4) The time constant of the circuit is :

5) The current in the inductor at $t= 500$ ms is:

6) The final value of the voltage $v(t= \infty)$ across the inductor is :

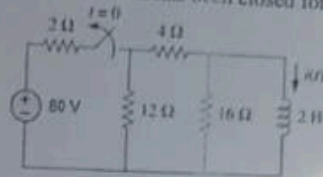


Fig.1

Prob. 3: For the questions 7,8,9,10 consider the circuit shown in figure 2 and answer:

7) The value of V_0 is:

8) The value of the current (i) in 5 Ω is:

9) The power absorbed by 3 Ω resistor is:

10) The power absorbed by 12 V voltage source is:

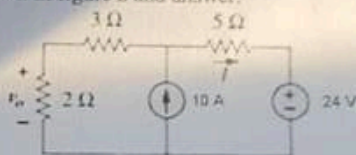


Fig.2

Prob. 4: Consider the circuit shown in figure 3. Use Thevenin's theorem with respect to terminals (a - b) and answer:

a. The Thevenin voltage is

b. The Thevenin resistance is.....

c. The current I is.....

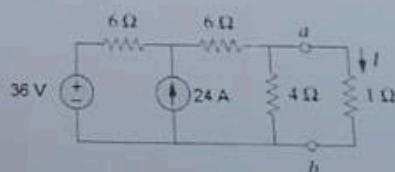


Fig. 3

d. What is the value of the load resistance that should be connected to the ab terminals to achieve maximum power transfer?

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a) Refer to figure 4; the value of the equivalent resistance is:

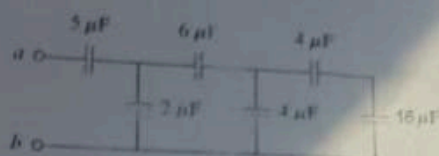


Fig. 4

b) Refer to figure 5; the value of the equivalent resistance is:

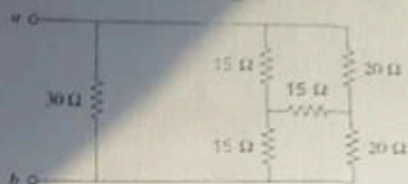


Fig. 5

Good Luck

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Problem 1: Inductor Analysis

Given: $L = 1 \text{ mH} = 10^{-3} \text{ H}$, $i(t) = 60 \cos(100t) \text{ mA} = 0.06 \cos(100t) \text{ A}$

1. **Voltage across the inductor $v(t)$:**

$$v(t) = L * (di/dt)$$

$$v(t) = 10^{-3} * d/dt [0.06 \cos(100t)]$$

$$v(t) = 10^{-3} * 0.06 * (-100 \sin(100t))$$

$$\mathbf{v(t) = -6 \sin(100t) \text{ Volts}}$$

2. **Energy stored $w(t)$:**

$$w(t) = 0.5 * L * i(t)^2$$

$$w(t) = 0.5 * 10^{-3} * (0.06 \cos(100t))^2$$

$$w(t) = 0.5 * 10^{-3} * 0.0036 \cos^2(100t)$$

$$\mathbf{w(t) = 1.8 \cos^2(100t) \text{ mJ}}$$

Problem 2: Transient Circuit (Fig. 1)

Requirement	Calculation / Result
Initial current $i(0+)$	The inductor acts as a short circuit at $t < 0$. $i(0-) = 60\text{V} / (2\Omega + 12\Omega) = 4.286 \text{ A}$. Therefore, $i(0+) = 4.286 \text{ A}$.
Time constant (τ)	$R_{th} = (2+12) \parallel 16 = 14 \parallel 16 = 7.467 \Omega$. $\tau = L/R_{th} = 0.002 / 7.467 \approx 0.268 \text{ s}$.
Current at $t=500\text{ms}$	$i(t) = i(0) * e^{-t/\tau} = 4.286 * e^{-0.5 / 0.268} \approx 0.66 \text{ A}$.
Final voltage $v(\infty)$	At steady state ($t = \infty$), the current is constant, hence $v = L(di/dt) = 0 \text{ V}$.

Problem 3: Circuit Analysis (Fig. 2)

To find V_0 and the current in the 5Ω resistor, apply Nodal Analysis:

- Define node voltages.
- Use Kirchhoff's Current Law (KCL) at the nodes.
- The power absorbed by the 3Ω resistor is $P = I^2 * R$.
- The power absorbed by the $12V$ source is $P = V * I$.

Problem 4: Thevenin's Theorem (Fig. 3)

1. **Thevenin Voltage (V_{th}):** Determine the open-circuit voltage at terminals a-b.
2. **Thevenin Resistance (R_{th}):** Equivalent resistance at terminals a-b after replacing independent sources with short/open circuits.
3. **Current I:** Calculate using $I = V_{th} / (R_{th} + 1\Omega)$.
4. **Maximum Power Transfer:** The load resistance R_{load} must equal R_{th} .

Problem (a): Equivalent Capacitance (Fig. 4)

Step-by-Step Calculation:

- 1. Parallel Combination (Right side):** The $4\ \mu\text{F}$ and $16\ \mu\text{F}$ capacitors are in parallel.
 $C_{p1} = 4\ \mu\text{F} + 16\ \mu\text{F} = 20\ \mu\text{F}$
- 2. Series Combination:** The $20\ \mu\text{F}$ equivalent is in series with the $6\ \mu\text{F}$ capacitor.
 $C_{s1} = (20 * 6) / (20 + 6) = 120 / 26 \approx 4.615\ \mu\text{F}$
- 3. Parallel Combination (Middle):** This $4.615\ \mu\text{F}$ is in parallel with the $2\ \mu\text{F}$ capacitor.
 $C_{p2} = 4.615\ \mu\text{F} + 2\ \mu\text{F} = 6.615\ \mu\text{F}$
- 4. Final Series Combination:** This is in series with the $5\ \mu\text{F}$ capacitor connected to terminals a-b.
 $C_{eq} = (6.615 * 5) / (6.615 + 5) = 33.075 / 11.615 \approx 2.848\ \mu\text{F}$

Problem (b): Equivalent Resistance (Fig. 5)

Analysis:

To find the equivalent resistance between terminals a and b, we simplify the right-hand network using series and parallel combinations:

- 1. Right Branch Series:** The top $15\ \Omega$ and $30\ \Omega$ resistors are in series, and the bottom $15\ \Omega$ and $20\ \Omega$ resistors are in series.
 $R_{top} = 15 + 30 = 45\ \Omega$
 $R_{bottom} = 15 + 20 = 35\ \Omega$
- 2. Parallel Reduction:** These two branches are in parallel with each other.
 $R_{right} = (45 * 35) / (45 + 35) = 1575 / 80 = 19.6875\ \Omega$
- 3. Total Resistance:** The $30\ \Omega$ resistor is in parallel with this entire right-hand block.
 $R_{eq} = (30 * 19.6875) / (30 + 19.6875) = 590.625 / 49.6875 \approx 11.88\ \Omega$